03_DE_plots

Libraries

Directories

▼ Code

```
base_dir <- "/nemo/lab/destrooperb/home/shared/zanettc/millie_pr
run_num <- "run1"

results_dir <- file.path(base_dir, "results", run_num)
objects_dir <- file.path(base_dir, "data", "processed", run_num)
raw_dir <- file.path(base_dir, "data", "raw")

dir.create(results_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(objects_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(raw_dir, recursive = TRUE, showWarnings = FALSE)
```

Read in MSstats output

▼ Code

```
res_df <- fread(file.path(results_dir, "Full_GroupComparison_ es
```

Mapping protein IDs to gene names

▼ Code

```
clean_spec_raw <- fread(file.path(objects_dir, "clean_spec_ra .c

protein_dictionary <- clean_spec_raw %>%
  select(Protein = PG.ProteinGroups,
         Gene = PG.Genes,
         Description = PG.ProteinDescriptions) %>%
  distinct()
```

Volcano function

Boundary cut-offs of 0.05 FDR and +/- 0.58 Log2FC (1.5x).

▼ Code

```
# Define colours
volcano_cols <- c("Upregulated" = "firebrick3",
                  "Downregulated" = "navy",
                  "Not Significant" = "grey80")

generate_swanky_volcano <- function(msstats_results, dict, compa

  # --- Data Processing ---
  volcano_data <- msstats_results %>%
    # Isolate the specific comparison and drop missing values
    filter(Label == comparison_label, !is.na(adj.pvalue), is.fin
    # Map the Gene names from the dictionary
    left_join(dict, by = "Protein") %>%
    mutate(
      # Fallback to Protein ID if Gene name is missing
      Gene = ifelse(is.na(Gene) | Gene == "", Protein, Gene),
      # Define significance categories
      Status = case_when(
        adj.pvalue < 0.05 & log2FC > logfc_cutoff ~ "Upregulated
        adj.pvalue < 0.05 & log2FC < -logfc_cutoff ~ "Downregula
        TRUE ~ "Not Significant"
      )
    ) %>%
    # Group by Status to rank Upregulated and Downregulated sepa
    group_by(Status) %>%
    # Rank by FDR (smallest adj.pvalue = 1)
    mutate(GroupRank = rank(adj.pvalue, ties.method = "first"))
    ungroup() %>%
    mutate(
      # Label top 20 for each significant category
      PlotLabel = case_when(
        Status != "Not Significant" & GroupRank <= 20 ~ Gene,
        TRUE ~ NA_character_
      )
    )

  # --- Plotting ---
  p <- ggplot(volcano_data, aes(x = log2FC, y = -log10(adj.pvalu
    geom_point(aes(size = Status, alpha = Status), shape = 21, s
    scale_fill_manual(values = volcano_cols) +
    scale_color_manual(values = c("Upregulated" = "black", "Down
```

```
    scale_size_manual(values = c("Upregulated" = 3, "Downregulat
    scale_alpha_manual(values = c("Upregulated" = 1, "Downregula
    geom_hline(yintercept = -log10(0.05), linetype = "dashed", c
    geom_vline(xintercept = c(-logfc_cutoff, logfc_cutoff), line
    geom_label_repel(
      aes(label = PlotLabel),
      fill = "white", color = "black", fontface = "bold", size =
      box.padding = 0.5, point.padding = 0.5,
      min.segment.length = 0, segment.color = "grey50", max.over
    ) +
    labs(
      title = paste0("Differential Abundance: ", gsub("_", " ",
      subtitle = paste0("Top 20 Genes Labeled | FDR < 0.05 | |lo
      x = bquote(bold("Log"[2] ~ "Fold Change")),
      y = bquote(bold("-Log"[10] ~ "FDR"))
    ) +
    theme_minimal(base_size = 14) +
    theme(
      plot.title = element_text(face = "bold", size = 18, hjust
      plot.subtitle = element_text(size = 12, color = "grey40",
      axis.title = element_text(face = "bold", size = 14),
      axis.text = element_text(face = "bold", color = "black"),
      axis.line = element_line(color = "black", linewidth = 1),
      axis.ticks = element_line(color = "black", linewidth = 1),
      legend.position = "top",
      legend.title = element_blank(),
      legend.text = element_text(size = 12, face = "bold"),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "grey95")
    )

  # --- Output Generation ---
  # Ensure directory exists
  dir.create(graphs_dir, showWarnings = FALSE, recursive = TRUE)
  safe_name <- gsub(" ", "_", comparison_label)

  # Save Plot
  ggsave(
    filename = file.path(graphs_dir, paste0("Volcano_", safe_nam
    plot = p,
    width = 10, height = 7, dpi = 300,
    device = cairo_pdf
  )

  # Save Data Table
  # Select only the most useful columns and sort by significance
  export_data <- volcano_data %>%
    select(Protein, Gene, Description, log2FC, pvalue, adj.pvalu
```

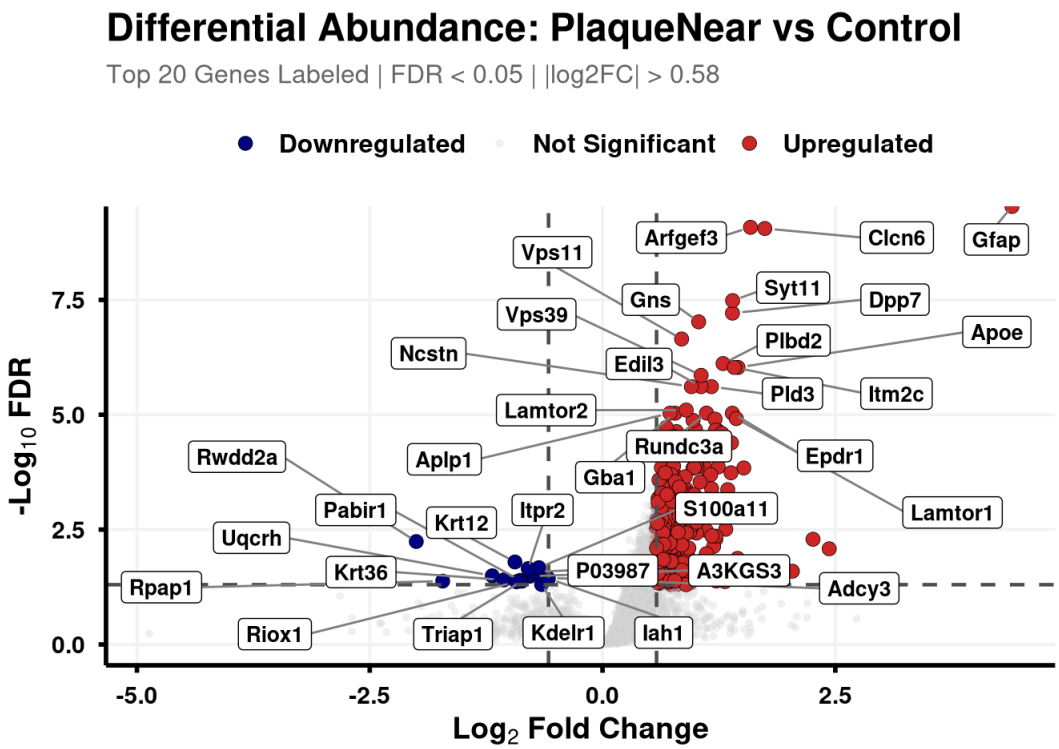
```
    arrange(adj.pvalue)

  fwrite(export_data, file = file.path(graphs_dir, paste0("Volca

  return(p)
}
```

Iterate through comparisons to generate plots and useful DE tables

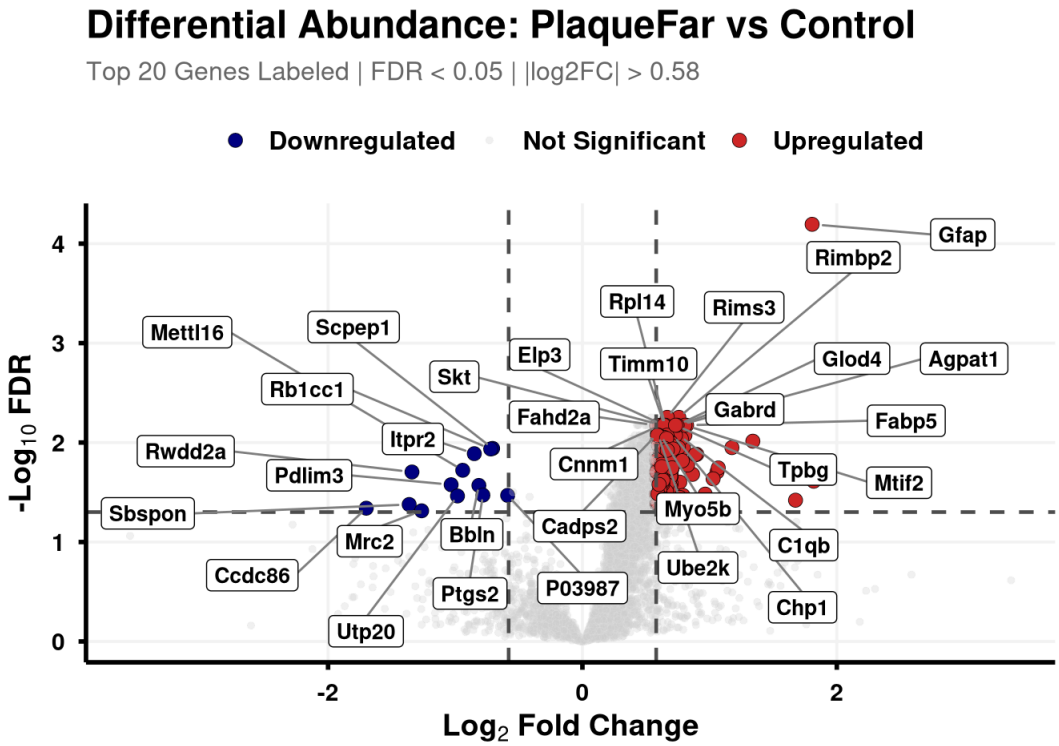
▼ Code

```
all_comparisons <- unique(res_df$Label)

# 2. Loop through each comparison, generate the plot/data, and s
plot_list <- list()

for (comp in all_comparisons) {
  p <- generate_swanky_volcano(
    msstats_results = res_df,
    dict = protein_dictionary,
    comparison_label = comp,
    logfc_cutoff = 0.58,
    graphs_dir = results_dir
  )

  # Store the plot in our list
  plot_list[[comp]] <- p

  # Print the plot so it appears in HTML output
  print(p)
}
```

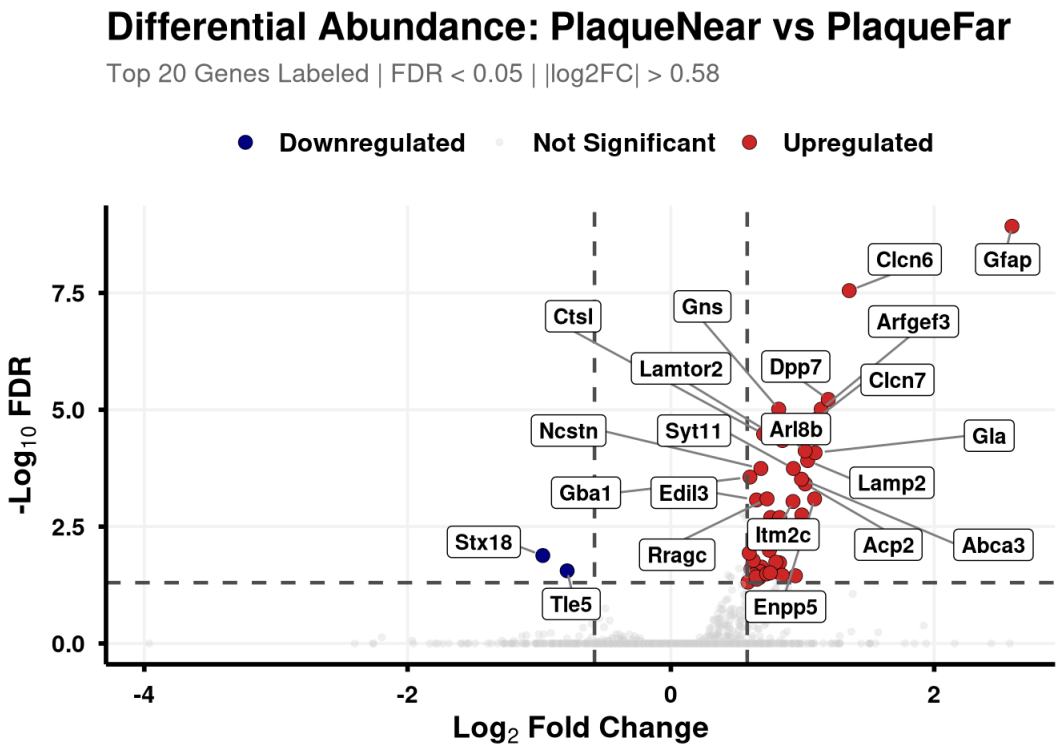
```
Warning: Removed 6846 rows containing missing values or values
outside the scale range
(`geom_label_repel()`).
Removed 6846 rows containing missing values or values outside
the scale range
(`geom_label_repel()`).

Warning: Removed 6867 rows containing missing values or values
outside the scale range
(`geom_label_repel()`).
```




```
Warning: Removed 6867 rows containing missing values or values
outside the scale range
(`geom_label_repel()`).

Warning: Removed 6904 rows containing missing values or values
outside the scale range
(`geom_label_repel()`).
```



```
Warning: Removed 6904 rows containing missing values or values
outside the scale range
(`geom_label_repel()`).
```



Interrogation of floor genes from Near vs Far plaque volcano plots

▼ Code

```
res_df
```

Protein	Label	log2FC	
<chr>	<chr>	<dbl>	
A0A1W2P872	PlaqueNear_vs_Control	6.358717e-02	0
A0A1W2P872	PlaqueFar_vs_Control	3.411036e-01	0
A0A1W2P872	PlaqueNear_vs_PlaqueFar	-2.775164e-01	0
A0A7N9VSG0	PlaqueNear_vs_Control	4.860182e-02	0
A0A7N9VSG0	PlaqueFar_vs_Control	1.444918e-01	0
A0A7N9VSG0	PlaqueNear_vs_PlaqueFar	-9.588995e-02	0
A1L317	PlaqueNear_vs_Control	-5.029654e-01	0
A1L317	PlaqueFar_vs_Control	-4.068406e-01	0
A1L317	PlaqueNear_vs_PlaqueFar	-9.612489e-02	0
A1L3P4	PlaqueNear_vs_Control	7.944477e-01	0

1-10 of 10,000 rows | 1-4 ... Previous **1** 2 3 4 5 6 ... 1000 Next

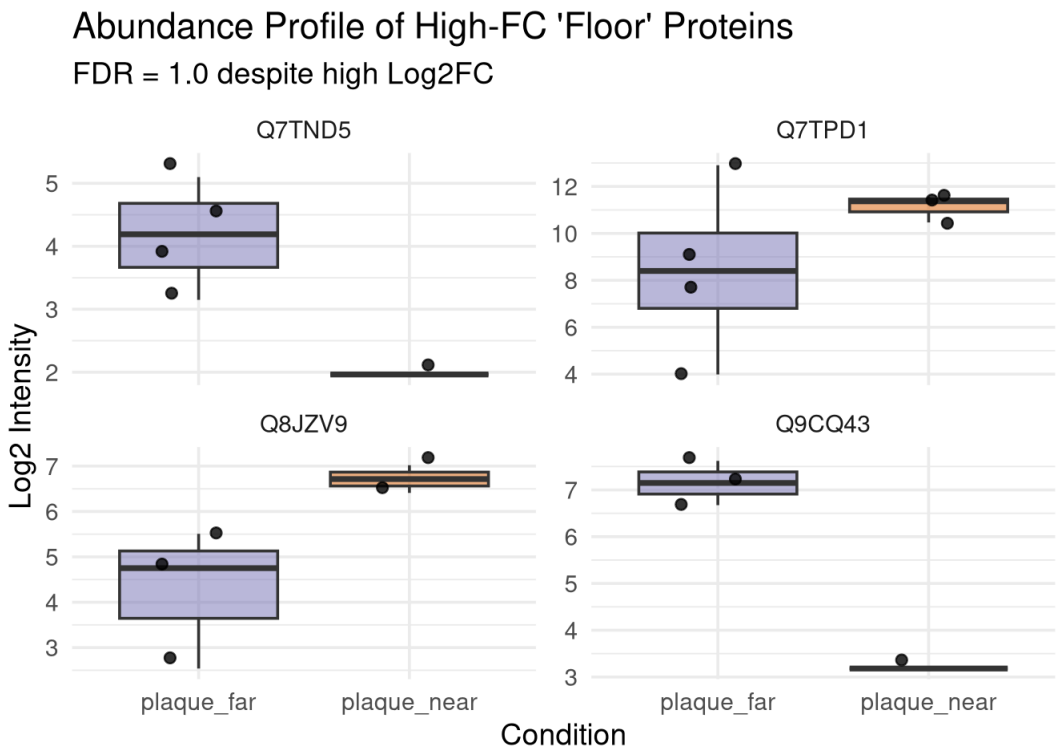
▼ Code

```
floor_check <- res_df %>%
  filter(Label == "PlaqueNear_vs_Control", adj.pvalue > 0.99) %>
  select(Protein, log2FC, pvalue, adj.pvalue, SE, DF) %>%
  arrange(desc(abs(log2FC)))

# Look at the top results
head(floor_check, 10)
```

Protein	log2FC	pvalue	adj.pvalue	SE	DF
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<in
Q148V8	-0.021676529	0.9893022	0.9946505	1.4326969	2
P58058	-0.021462339	0.9901955	0.9952578	1.6618851	5
Q80Y44	-0.007800151	0.9950497	0.9976595	1.0031009	1
P01942	-0.007053665	0.9918116	0.9962140	0.6847425	67
Q9JIH2	0.006547290	0.9953405	0.9976603	1.0329166	3
P16056	0.006516866	0.9964225	0.9981633	1.2880872	2
Q03145	0.005869501	0.9886924	0.9946188	0.3304192	1
Q921X6	0.005338701	0.9968483	0.9983582	1.2704376	4
Q8VID5	0.005172308	0.9866310	0.9928353	0.3066527	38
O88708	0.004843876	0.9935990	0.9969313	0.5889010	10

1-10 of 10 rows

▼ Code

```
#filtered for top 4 logFC proteins with >0.99 FDR
target_proteins <- res_df %>%
  filter(Label == "PlaqueNear_vs_PlaqueFar", adj.pvalue > 0.99)
  arrange(desc(abs(log2FC))) %>%
  slice_head(n = 4) %>%
  pull(Protein)

#read in msstats data
processed_data <- qs_read(file.path(objects_dir, "processed_msst

# extract their summarized abundances from the processed MSstats
abundance_data <- processed_data$ProteinLevelData %>%
  filter(Protein %in% target_proteins) %>%
  # Filter to just the two conditions of interest for clarity
  filter(GROUP %in% c("plaque_near", "plaque_far")) %>%
  group_by(Protein, GROUP, SUBJECT) %>%
  summarise(LogIntensities = mean(LogIntensities, na.rm = TRUE),
```

```
#plot abundances
#each dot represents the summarised abundance for a biological r
p_abundance <- ggplot(abundance_data, aes(x = GROUP, y = LogInte
  geom_boxplot(alpha = 0.5, outlier.shape = NA) +
  geom_jitter(width = 0.2, size = 2, alpha = 0.8) +
  facet_wrap(~ Protein, scales = "free_y") +
  theme_minimal(base_size = 14) +
  scale_fill_manual(values = c("plaque_near" = "#d95f02", "plaqu
  labs(
    title = "Abundance Profile of High-FC 'Floor' Proteins",
    subtitle = "FDR = 1.0 despite high Log2FC",
    x = "Condition",
    y = "Log2 Intensity"
  ) +
  theme(legend.position = "none")

# Print the plot
print(p_abundance)
```